STOR 557 HW 6

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```
library(faraway)
```

```
## Warning: package 'faraway' was built under R version 4.4.3
```

```
data(death)
head(death)
```

```
##     y penalty victim defend
## 1  19     yes      w      w
## 2 132      no      w      w
## 3   0     yes      b      w
## 4   9      no      b      w
## 5  11     yes      w      b
## 6  52      no      w      b
```

```
tab <- xtabs(y ~ defend + penalty, data = death)
tab
```

```
##       penalty
## defend  no yes
##      b 149  17
##      w 141  19
```

```
prop.table(tab, 1)
```

```
##       penalty
## defend        no       yes
##      b 0.8975904 0.1024096
##      w 0.8812500 0.1187500
```

```
tab2 <- xtabs(y ~ defend + penalty + victim, data = death)
round(prop.table(tab2, c(1, 3)) * 100, 2)
```

```
## , , victim = b
##
##       penalty
## defend     no    yes
##      b  94.17   5.83
```

```
##      w 100.00   0.00
##
## , , victim = w
##
##       penalty
## defend     no    yes
##      b  82.54  17.46
##      w  87.42  12.58
```

```
fisher.test(tab)
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  tab
## p-value = 0.7246
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##  0.5560159 2.5243470
## sample estimates:
## odds ratio
##   1.180455
```

```
fisher.test(tab2[, , "b"])
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  tab2[, , "b"]
## p-value = 1
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##   0.00000 10.72162
## sample estimates:
## odds ratio
##          0
```

```
fisher.test(tab2[, , "w"])
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  tab2[, , "w"]
## p-value = 0.3893
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##  0.2849356 1.7014713
## sample estimates:
## odds ratio
##  0.6817309
```

```r
mantelhaen.test(tab2)
```

```
##
##  Mantel-Haenszel chi-squared test with continuity correction
##
## data:  tab2
## Mantel-Haenszel X-squared = 0.79633, df = 1, p-value = 0.3722
## alternative hypothesis: true common odds ratio is not equal to 1
## 95 percent confidence interval:
##  0.2864136 1.4091583
## sample estimates:
## common odds ratio
##         0.6352968
```

Overall, White defendants received the death penalty more often than Black defendants (11.88% vs. 10.24%). However, when looking at the victim's race, Black defendants had higher rates in both groups. This shows Simpson's paradox. All p-values were greater than 0.05, so none of the differences were statistically significant.

```r
m1 <- glm(y ~ defend + victim + penalty,
          family = poisson, data = death)

m2 <- glm(y ~ defend*victim + victim*penalty,
          family = poisson, data = death)

m3 <- glm(y ~ (defend + victim + penalty)^2,
          family = poisson, data = death)

AIC(m1, m2, m3)
```

```
##    df       AIC
## m1  4 181.61094
## m2  6  49.56348
## m3  7  50.38232
```

```r
anova(m2, m3, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: y ~ defend * victim + victim * penalty
## Model 2: y ~ (defend + victim + penalty)^2
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         2    1.88190
## 2         1    0.70074  1   1.1812   0.2771
```

```r
anova(m1, m2, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: y ~ defend + victim + penalty
## Model 2: y ~ defend * victim + victim * penalty
```

```
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1         4    137.929
## 2         2      1.882  2   136.05 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Model 2 is the most appropriate model. It includes interactions between defendant race and victim race, and between victim race and the death penalty. Model 2 fits significantly better than Model 1 ($p < 0.001$), while Model 3 does not significantly improve the fit ($p = 0.2771$).

```
b1 <- glm(I(penalty == "yes") ~ victim,
          weights = y, family = binomial,
          data = death)

b2 <- glm(I(penalty == "yes") ~ victim + defend,
          weights = y, family = binomial,
          data = death)

summary(b1)
```

```
##
## Call:
## glm(formula = I(penalty == "yes") ~ victim, family = binomial,
##     data = death, weights = y)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -2.8717     0.4196  -6.843 7.75e-12 ***
## victimw       1.0579     0.4635   2.282   0.0225 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 226.51  on 6  degrees of freedom
## Residual deviance: 220.26  on 5  degrees of freedom
## AIC: 224.26
##
## Number of Fisher Scoring iterations: 5
```

```
anova(b1, b2, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: I(penalty == "yes") ~ victim
## Model 2: I(penalty == "yes") ~ victim + defend
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         5     220.26
## 2         4     219.08  1   1.1812   0.2771
```

The binomial regression shows that victim race is significant ($p = 0.0225$), but adding defendant race does not significantly improve the model ($p = 0.2771$). Therefore, the model using only victim race is preferred. This agrees with the dependence model from part (b).

```r
library(faraway)
data(uncviet)
head(uncviet)
```

```
##     y policy  sex  year
## 1 175      A Male Fresh
## 2 116      B Male Fresh
## 3 131      C Male Fresh
## 4  17      D Male Fresh
## 5 160      A Male  Soph
## 6 126      B Male  Soph
```

```r
str(uncviet)
```

```
## 'data.frame':    40 obs. of  4 variables:
##  $ y     : num  175 116 131 17 160 126 135 21 132 120 ...
##  $ policy: Factor w/ 4 levels "A","B","C","D": 1 2 3 4 1 2 3 4 1 2 ...
##  $ sex   : Factor w/ 2 levels "Female","Male": 2 2 2 2 2 2 2 2 2 2 ...
##  $ year  : Factor w/ 5 levels "Fresh","Grad",..: 1 1 1 1 5 5 5 5 3 3 ...
```

```r
m1 <- glm(y ~ sex + year + policy,
          family = poisson, data = uncviet)
m2 <- glm(y ~ (sex + year + policy)^2,
          family = poisson, data = uncviet)

anova(m1, m2, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: y ~ sex + year + policy
## Model 2: y ~ (sex + year + policy)^2
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1        31     423.83
## 2        12      19.19 19   404.63 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
drop1(m2, test = "Chisq")
```

```
## Single term deletions
##
## Model:
## y ~ (sex + year + policy)^2
##             Df Deviance    AIC     LRT  Pr(>Chi)
## <none>           19.194 299.58
## sex:year     4   70.643 343.03  51.449 1.798e-10 ***
## sex:policy   3  153.935 428.32 134.741 < 2.2e-16 ***
## year:policy 12  216.312 472.70 197.118 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The model with all two-way interactions is the most appropriate. All three interactions (sex × year, sex × policy, and year × policy) are statistically significant ($p < 0.05$). The model also fits the data adequately ($p = 0.084$), so a three-way interaction is not necessary.

```
uncviet$year_score <- match(uncviet$year,
                            c("Fresh", "Soph", "Junior",
                              "Senior", "Grad"))

uncviet$policy_score <- match(uncviet$policy,
                              c("A", "B", "C", "D"))

m_score <- glm(y ~ sex*year + sex*policy +
                year_score:policy_score,
               family = poisson, data = uncviet)

summary(m_score)
```

```
##
## Call:
## glm(formula = y ~ sex * year + sex * policy + year_score:policy_score,
##     family = poisson, data = uncviet)
##
## Coefficients:
##                           Estimate Std. Error z value Pr(>|z|)
## (Intercept)                2.65367    0.15369  17.267  < 2e-16 ***
## sexMale                    2.41212    0.16160  14.926  < 2e-16 ***
## yearGrad                  -0.89963    0.19565  -4.598 4.26e-06 ***
## yearJunior                -0.08298    0.15243  -0.544  0.58619
## yearSenior                -1.04264    0.18252  -5.713 1.11e-08 ***
## yearSoph                  -0.85032    0.18427  -4.615 3.94e-06 ***
## policyB                   -0.14240    0.15850  -0.898  0.36896
## policyC                    0.52402    0.15564   3.367  0.00076 ***
## policyD                   -2.41807    0.24695  -9.792  < 2e-16 ***
## sexMale:yearGrad          -0.06674    0.15158  -0.440  0.65973
## sexMale:yearJunior        -0.64146    0.15466  -4.148 3.36e-05 ***
## sexMale:yearSenior        -0.00560    0.16739  -0.033  0.97331
## sexMale:yearSoph           0.51304    0.19398   2.645  0.00818 **
## sexMale:policyB           -0.51588    0.16340  -3.157  0.00159 **
## sexMale:policyC           -1.35581    0.14050  -9.650  < 2e-16 ***
## sexMale:policyD           -0.37927    0.21797  -1.740  0.08185 .
## year_score:policy_score    0.17220    0.01364  12.624  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 2708.080  on 39  degrees of freedom
## Residual deviance:   46.967  on 23  degrees of freedom
## AIC: 305.35
##
## Number of Fisher Scoring iterations: 4
```

```r
uncviet$year_score2 <- c(Fresh=1, Soph=2, Junior=3,
                         Senior=4, Grad=6)[as.character(uncviet$year)]

uncviet$policy_score2 <- c(A=1, B=2, C=3,
                           D=5)[as.character(uncviet$policy)]

m_alt <- glm(y ~ sex*year + sex*policy +
              year_score2:policy_score2,
             family = poisson, data = uncviet)

summary(m_alt)
```

```
##
## Call:
## glm(formula = y ~ sex * year + sex * policy + year_score2:policy_score2,
##     family = poisson, data = uncviet)
##
## Coefficients:
##                             Estimate Std. Error z value Pr(>|z|)
## (Intercept)                 2.602433   0.154830  16.808  < 2e-16 ***
## sexMale                     2.455591   0.163097  15.056  < 2e-16 ***
## yearGrad                   -0.731389   0.184109  -3.973 7.11e-05 ***
## yearJunior                  0.161489   0.144907   1.114  0.26510
## yearSenior                 -0.665009   0.166647  -3.991 6.59e-05 ***
## yearSoph                   -0.732366   0.182913  -4.004 6.23e-05 ***
## policyB                    -0.014777   0.156301  -0.095  0.92468
## policyC                     0.792534   0.144320   5.492 3.99e-08 ***
## policyD                    -2.536451   0.253265 -10.015  < 2e-16 ***
## sexMale:yearGrad           -0.121298   0.150975  -0.803  0.42172
## sexMale:yearJunior         -0.685425   0.154077  -4.449 8.64e-06 ***
## sexMale:yearSenior         -0.067459   0.166394  -0.405  0.68517
## sexMale:yearSoph            0.490026   0.193849   2.528  0.01148 *
## sexMale:policyB            -0.518505   0.163255  -3.176  0.00149 **
## sexMale:policyC            -1.358984   0.139987  -9.708  < 2e-16 ***
## sexMale:policyD            -0.380578   0.218884  -1.739  0.08208 .
## year_score2:policy_score2   0.121056   0.009271  13.057  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 2708.08  on 39  degrees of freedom
## Residual deviance:   30.42  on 23  degrees of freedom
## AIC: 288.81
##
## Number of Fisher Scoring iterations: 4
```

```r
AIC(m_score, m_alt, m2)
```

```
##         df      AIC
## m_score 17 305.3537
## m_alt   17 288.8063
## m2      28 299.5806
```

The model shows a significant positive relationship between year and opinion ($p < 0.001$). Students in higher school years tend to favor less military involvement in Vietnam. Changing the scores did not change this conclusion. The alternative scores gave a better fit, with a lower AIC of 288.81 compared to 305.35.